

A systematic review and meta-analysis of predilation and postdilation in transfemoral carotid artery stenting

Behrad Ziapour, MD,^a Marc L. Schermerhorn, MD,^b Mark D. Iafrati, MD,^a Luis B. Suarez, MD,^a Shahab TourSavadkahi, MD,^c and Payam Salehi, MD, PhD,^a *Boston, Mass; and Baltimore, Md*

ABSTRACT

Objective: In this review, we evaluate the short-term impact of different ballooning combinations on hemodynamic instability, neurologic events, and mortality associated with carotid artery stenting.

Methods: We used a search strategy to identify studies in the Cochrane Library trials register, PubMed, EMBASE, Google Scholar, and the US National Library of Medicine clinical trial databases up to May 2019. Studies reporting either hemodynamic instability, neurologic events, or mortality were included according to the recommendations of the Preferred Reporting Items for Systematic reviews and Meta-analysis. Fixed and random models were used to summarize the effects.

Results: Of seven included cohort studies for qualitative review, six were selected for meta-analysis. Patients without postdilation had significantly less hemodynamic instability in the random effects model (risk ratio, 0.53; 95% confidence interval, 0.37-0.77; $P = .001$). Regardless of the dilation type, the overall summary effect in the fixed model showed a significant decrease in the risk of developing short-term neurologic events (risk ratio, 0.67; 95% confidence interval, 0.47-0.97; $P = .03$). Last, the exclusion of either of the two dilation techniques had no significant effect on mortality.

Conclusions: Avoiding postdilation reduces perioperative hemodynamic instability and this benefit lasts at least up to 30 days. Regardless of the type, fewer dilations during carotid artery stenting possibly decreases the neurologic events during and 30 days after the procedure. The data also suggest that the residual stenoses above current practiced standard may provide a safer threshold to prompt postdilation. Statements about the long-term effects of dilation techniques requires more study. (*J Vasc Surg* 2020;■:1-11.)

Keywords: carotid artery stenosis; Carotid stenting; Predilation; Postdilation

Carotid endarterectomy (CEA) combined with best medical treatment is the gold standard of care in patients with carotid artery stenosis who require intervention.^{1,2} However, with the introduction of embolic protection devices (EPDs), refined techniques, improved patient selection, and more experienced operators, periprocedural complications in CAS, including stroke or death have considerably decreased.³ Therefore, CAS is still being used as an alternative to CEA, especially in patients with high surgical risk.

Angioplasty has been used to prepare a stenosed vessel for stent placement (predilation), or to mitigate any residual stenosis after deployment (postdilation). Several

studies have shown the effect of postdilation on hemodynamic stability and long-term in-stent restenosis,⁴⁻⁶ and a few studies have published the effect of predilation or postdilation on mortality and neurologic events. To date, the effect of predilation, postdilation, dual dilation, or no dilation during carotid artery stenting remains inconclusive. Therefore, in this review, we evaluate the short-term impact of dilation methods on hemodynamic instability rates, neurologic events, and mortality associated with transfemoral CAS.

METHODS

Literature search and search strategy. The review protocol is registered in PROSPERO (ID: CRD42019132314). We systematically searched the Cochrane Library trials register, PubMed, EMBASE, Google Scholar, and the US National Library of Medicine clinical trial databases for randomized controlled trials, cohort studies, and case-control studies that included any terms for carotids stenosis and terms related to predilation and postdilation from inception to May 2019. There was no language restriction; all studies were limited to humans. Our core search consisted of terms related to predilation or postdilation, combined with specific terms for carotid stenosis ([Appendix](#), online only).

Inclusion and exclusion criteria. Studies regarding radiation-induced stenosis were not excluded, but considered as a separate group for study, because the

From the Division of Vascular Surgery, Cardiovascular Center, Tufts Medical Center,^a and the Division of Vascular and Endovascular Surgery, Beth Israel Deaconess Medical Center,^b Boston; and the Aortic Center, Division of Vascular Surgery, Department of Surgery, University of Maryland Medical Center & School of Medicine, Baltimore.^c

Author conflict of interest: none.

Additional material for this article may be found online at www.jvascsurg.org.

Correspondence: Payam Salehi, MD, PhD, Assistant Professor of Surgery, Vascular Surgery, Tufts Medical Center, Tufts University School of Medicine, 800 Washington St #1035, Boston, MA 02111 (e-mail: psalehi@tuftsmedicalcenter.org).

The editors and reviewers of this article have no relevant financial relationships to disclose per the JVS policy that requires reviewers to decline review of any manuscript for which they may have a conflict of interest.

0741-5214

Copyright © 2019 by the Society for Vascular Surgery. Published by Elsevier Inc.

<https://doi.org/10.1016/j.jvs.2019.11.044>

etiology of stenosis in radiation-induced carotid disease remains unclear; therefore, plaque stability and behavior can be different. Two evaluators independently applied the pilot inclusion/exclusion criteria for the study selection process to a subset of studies from the bibliographies of candidate studies. Based on the results, we examined points of disagreement to provide an acceptable level of reliability during the process. Studies were excluded if they presented the overall outcome of a mixed group of patients with predilation and postdilation. All included studies were required to report either mortality, neurologic events, or hemodynamic instability. Mortality was defined as death within 30 days of the procedure. Included studies considered patients qualified to receive transfemoral CAS if they had asymptomatic carotid artery stenosis of greater than 70% or stenosis of greater than 50% with a history of cerebrovascular events and/or critical stenosis of the contralateral carotid artery. Neurologic events were defined as clinically confirmed transient ischemic attacks or strokes that occurred during the perioperative period or within 30 days after procedure. This definition excluded asymptomatic neuroimaging lesions. Hemodynamic instabilities included any of following changes without evidence of bleeding, hypovolemia, or heart failure during procedure or within 30 days from procedure. An absolute systolic blood pressure of less than 90 mm Hg or a mean arterial pressure of less than 50 mm Hg and/a or heart rate of less than 60 bpm or a proportional decrease (40% for systolic blood pressure, 20% for heart rate) in their initial values. Studies were not excluded owing to the use of routine preprocedural atropine.

Data extraction. Two authors (BZ and PS) independently identified data from the included studies. They used a data extraction form to collect data on study characteristics (author, year, design, sample size), patients' demographics (age, sex, EPDs use, percentage of symptomatic/asymptomatic included patients, stenosis mean), the number of patients in each intervention group (with or without one or both dilation techniques), and outcomes (mortality, neurologic events, and hemodynamic instability).

Assessment of quality and risk of bias. Two authors independently evaluated the quality of the retrieved studies according to the Newcastle-Ottawa scale. The retrieved papers were evaluated based on the selection of cohorts (0-4 points), comparability of cohorts (0-2 points), and exposure/outcomes of the participants (0-3 points). Studies with 7 to 9 points were considered high quality.

The quality of the retrieved observational articles was assessed using the Newcastle-Ottawa Scale. As recommended,⁷ to increase the sample size and provide

greater precision of the estimated effects and to avoid limitations of the threshold approach, we did not exclude any studies solely based on quality. Instead, we deferred this decision until conducting a sensitivity analysis. Furthermore, the limitations of these studies are discussed in this review.

Statistical analysis. This study was performed according to the recommendations of the Preferred Reporting Items for Systematic reviews and Meta-analysis (PRISMA) statement.⁸ Stata version 14.0 (Stata Corporation, College Station, Tex) was used for the meta-analysis. Dichotomous outcomes (including the mortality rates, neurologic events, and hemodynamic instability) were summarized by the risk ratio (RR) and 95% confidence interval (CI), using both random and fixed effect models. The results were considered statistically significant at a *P* value of less than .05 if the 95% CI did not include the value of 1. An inverse variance random effects model was used to conduct the meta-analysis. Also, an inverse variance fixed effects model was repeated to compare the results if there was a moderate heterogeneity (ie, $I^2 = 30\%-60\%$) or a heterogeneity level with little importance (ie, $I^2 < 40\%$).⁹ Two studies specifically had data regarding less intensive predilation models while the exposed and the nonexposed groups were matched regarding postdilation in both studies. As applies to this specific example, when results of a smaller study are systematically different from results of larger ones, which can happen as a result of publication bias or within-study bias in smaller studies,^{10,11} then a random effects meta-analysis will exacerbate the effects of the bias. Therefore, we only used a fixed model to compare the neurologic events of less intensive predilation models with the standard technique.

According to Higgins's theory, heterogeneity was explored using the χ^2 test.⁹ The quantity of heterogeneity was measured by the I^2 statistic¹² and shown in the forest plot. In addition to assessing the I^2 statistic, we examined the forest plot to visually assess heterogeneity.

A univariate metaregression analysis was performed for using EPDs, symptomatic lesions, and sex to determine whether these factors modify the association between dilations used during CAS and outcomes at the study level and whether they contribute to the heterogeneity among studies.¹³ We used a Knapp-Hartung modification model for this analysis. We also conducted an influence-analysis to assess the effect of each study on the summary risk estimate and their 95% CIs.¹⁴

RESULTS

Study selection. The initial search yielded 2288 potentially relevant studies in addition to 56 studies from a hand search and 150 studies from Google scholar. Of these, 301 were excluded as duplicate studies and 2003 were excluded based on the titles and abstracts. Therefore, the

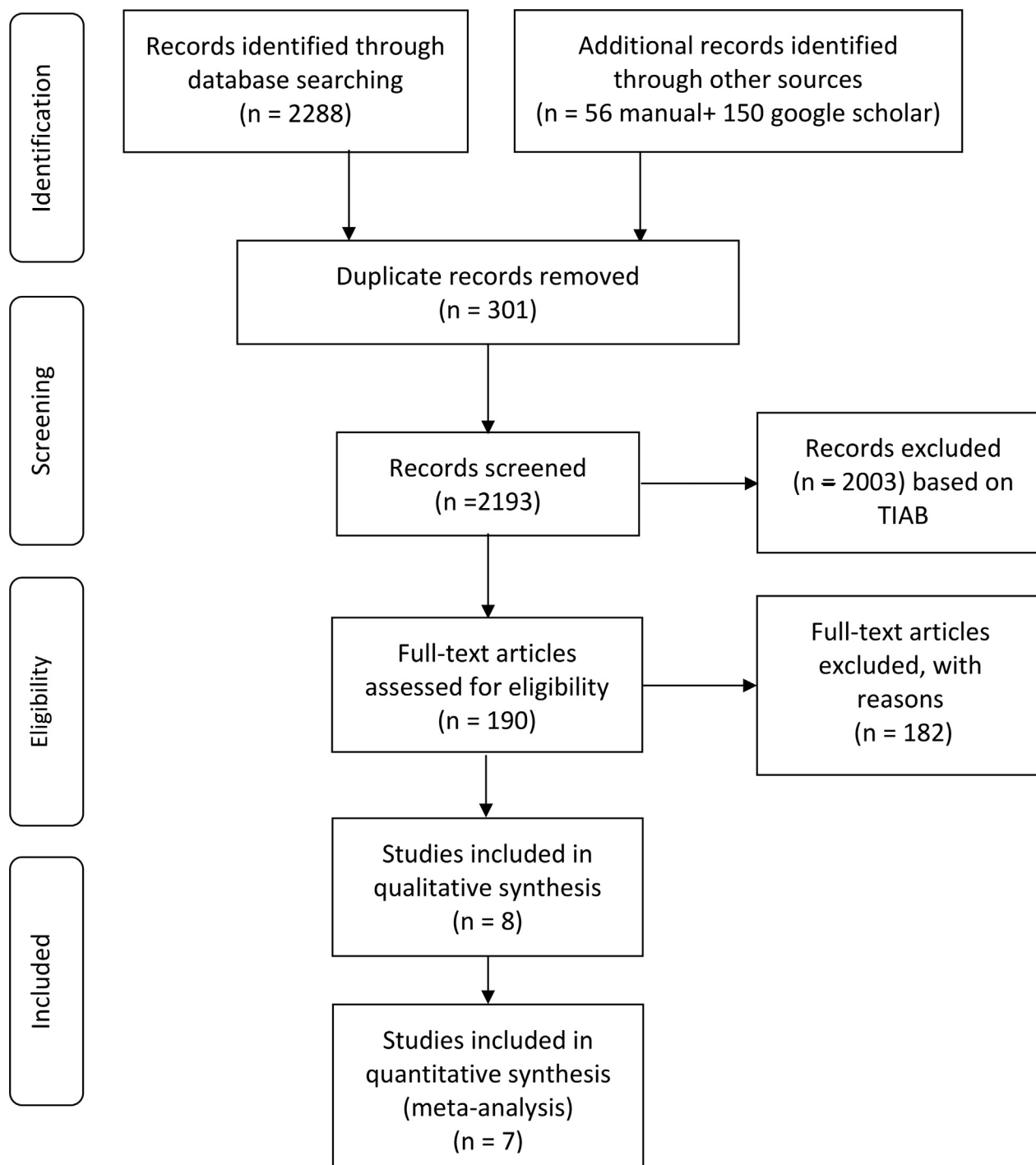


Fig 1. Flow chart of the identified, included and excluded studies as recommended by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). *TIAB*, Title and abstract

full texts of 190 studies were read for further evaluation. One hundred eighty-two studies were excluded mainly because they either had insufficient data on patients' dilation status or reported a mixed outcome of patients' dilation with and without dilations of interest ([Appendix](#), online only). Consequently, 8 studies were included in our systematic review with data from 4652 carotids stented.¹⁵⁻²¹ Of

these eight studies, six had compared at least an outcome from carotids stented with both dilations (exposed group) with carotids stented with only one dilation (nonexposed group). Two studies had data comparing the standard predilation with a less intensive predilation model. Both studies defined less intensive predilation as inflating the balloon more than 3 mm but less than 5 mm.^{20,21} However,

Table I. Characteristics of included studies

Author/year	Design	Country	Male, %	Symptomatic, %	EPD, %	Matched factors ^a	No. of carotids stented	No. of patients with dilations		
								Pre + Post –	Pre – Post +	Pre + Post +
Obeid et al, 2015 ¹⁵	Retrospective	USA	63.3	35.9	100	2, 3, 5, 7	3772	343	1446	1983
Li et al, 2005 ¹⁶	Prospective	China	61.3	93.4	0	9	270	201	63	6
Sahin et al, 2013 ¹⁸	Prospective	Turkey	75	44	100	1, 2, 4, 6, 7, 8, 9	275	–	233 ^b	42 ^b
Qazi et al, 2014 ¹⁹	Retrospective	USA	49.5	33	100	1, 2, 4, 5, 6, 8	103	31	–	72
Jin et al, 2010 ²¹	Prospective	Korea	–	72	100	3, 4, 9	99	52	–	47
Han et al, 2015 ¹⁷	Prospective	China	79	–	100	9	71	66	–	5
Sadato et al, 2004 ²⁰	Prospective	Japan	82	39.5	88	9	43	–	35	8
Harrod-Kim et al, 2005 ²²	Prospective	USA	69	25	0	10	19	8	2	9

EPD, Embolus protection device; Pre +/–, with/without predilation; Post +/–, with/without postdilation.

^aMatched factors are (1) age, (2) sex, (3) stenosis percentage before procedure, (4) symptomatic-patients percentage, (5) using β -blocker, (6) history of hypertension, (7) history of diabetes mellitus, (8) history of coronary artery disease, (9) prophylactic use of atropine, (10) radiation-induced stenosis.

^bThe original study reported that 89% and 95% of patients were respectively postdilated in groups with and without predilation (P value = .21). Therefore, data were included (under postdilated) to calculate the pooled effect of predilation on outcomes.

standard predilation inflates angioplasty balloon greater than 5 mm. Data regarding this optional model were analyzed independently. Also, a single study had reported neurologic outcomes after stenting¹⁹ radiation-induced carotid stenoses cases.²² This study was individually analyzed. Fig 1 shows the flow chart for our search strategy.

Characteristics, quality, and bias assessment of the included studies. Of the identified studies (Table I), two were retrospective cohort studies and five were prospective cohort studies. The number of stented carotids in these studies ranged from 19 to 3772. All eight studies included male and female adults with either symptomatic or asymptomatic carotid stenosis. The percentage of male participants varied from 25% to 82%, and the percentage of symptomatic lesions varied widely among studies from 33.0% to 93.4%. One study had not reported male percentage,²¹ and another study had not reported the percentage of symptomatic lesions.¹⁷ Participants with EPDs in one study did not have sufficient data regarding dilation status.¹⁶ Thus, only data of carotids without EPDs from that study were included in the final analysis. The study by Sadato et al²⁰ was excluded from the meta-analysis because they had only compared the standard predilation with their less intensive suggested predilation model. Only 88% of stented carotids in their study were protected by EPDs. All participants of five other studies had EPDs before stent deployment. One study had performed a logistic regression analysis of the effect of different dilation combinations.¹⁵ We only used their crude data originally extracted from the Vascular Quality Initiative database between 2005 and 2014. In an included study, 89% and 95% of patients were

respectively postdilated in groups with and without predilation ($P = .21$).¹⁸ Because the intervention and control groups were matched regarding predilation distribution, data from this study were included to analyze the impact of postdilation on outcomes under review. Of five studies included for hemodynamic instability analysis, two never used prophylactic atropine,^{19,21} and in the other, 0.5 to 1.0 mg of atropine was routinely administered intravenously before balloon inflation in all intervention and control groups patients.¹⁸ Participants prophylactic atropine data were not provided in the third study,¹⁵ and only 18% of all carotids in the last study were prophylactically treated with atropine.¹⁷ However, they showed in this last article that patients did not develop hemodynamic instabilities at significantly different rates relative to their atropine prophylaxis status ($P = .6$).

The risk of bias is summarized in Table II. Based on the selection, compatibility, and outcome categories presented by the Newcastle-Ottawa Quality Assessment Form for Cohort Studies, four studies^{15,18,19,21} were considered good quality studies and three studies^{16,17,20} were considered of poor quality. These latter studies were mostly considered as poor quality because they had not controlled factors between our intervention groups of interest.

Primary outcome: Short-term hemodynamic instability. Five studies reported the perioperative hemodynamic instability events during or after 4310 carotid stenting.^{15,17-19,21} The pooled analysis (Fig 2) indicated that the short-term hemodynamic instabilities were significantly less frequent when carotids were not postdilated compared with carotids that were postdilated. In

Table II. Assessment the risk of bias of studies

Author/year	Selection	Comparability	Outcomes	Overall quality
Obeid et al, 2015 ¹⁵	4	2	3	Good
Li et al, 2005 ¹⁶	4	0	3	Poor
Sahin et al, 2013 ¹⁸	4	2	3	Good
Qazi et al, 2014 ¹⁹	4	2	3	Good
Jin et al, 2014 ²¹	4	2	3	Good
Han et al, 2015 ¹⁷	4	0	3	Poor
Sadato et al, 2004 ²⁰	4	0	3	Poor
Harrod-Kim et al, 2005 ²²	4	1	3	Good

Selection of cohorts (0-4 points), Comparability of cohorts (0-2 points), and exposure/outcomes of the participants (0-3 points). Good quality: 3 or 4 stars in selection domain AND 1 or 2 stars in comparability domain AND 2 or 3 stars in outcome/exposure domain. Fair quality: 2 stars in selection domain AND 1 or 2 stars in comparability domain AND 2 or 3 stars in outcome/exposure domain. Poor quality: 0 or 1 star in selection domain OR 0 stars in comparability domain OR 0 or 1 stars in outcome/exposure domain.

a random pooling model, the RR of hemodynamic instability events of nonpredilated carotids vs predilated carotids was 0.58 (95% CI, 0.16-2.08; $P = .07$), and the RR of developing hemodynamic instabilities in non-postdilated carotids compared with postdilated carotids was 0.59 (95% CI, 0.39-0.87; $P = .03$).

Secondary outcomes: Short-term neurologic events and mortality. The included studies also had data identifying neurologic events. The impact of dilation techniques on developing neurologic events was not significant in the random effects model (Fig 3). The RR estimates for using predilation or postdilation were 0.73 (95% CI, 0.34-1.54; $P = .4$) and 0.53 (95% CI, 0.11-2.57; $P = .43$), respectively. However, the overall summary effect of performing a single vs double dilations (regardless of the type) in the fixed model showed a significant decrease in the risk of developing short term neurologic events (RR, 0.67; 95% CI, 0.47-0.97; $P = .03$; Supplementary Figs 1 and 2, online only). Also, results revealed that a less intensive predilation causes significantly fewer neurologic events than standard technique (RR, 0.27; 95% CI, 0.09-0.86; $P = .026$; Fig 4). In radiation-induced carotid stenoses, avoiding predilation or postdilation did not cause a significantly different number of neurologic events than by standard technique. The RR of developing neurologic events were respectively 4.5 (95% CI, 0.26-1.2; $P = .2$) and 0.37 (95% CI, 0.02-7.99; $P = .52$) without predilation or postdilation.

The pooled effects of using predilation or postdilation on mortality in the random effects model were nonsignificant (Fig 5). Mortality RR estimates were 0.56 (95%

CI, 0.26-1.2; $P = .14$) and 1.05 (95% CI, 0.36-3.02; $P = .9$), respectively, for using predilation and postdilation in carotid stenting.

Publication bias. Funnel plots were used to assess publication bias. The visible asymmetric distribution of the studies most likely indicates the presence of some publication bias or systematic heterogeneity regarding hemodynamic instabilities (Fig 6), but not for the other outcomes of this review.

Metaregression. Potential confounders, reported by most of the studies, were symptomatic lesions, using EPDs, and sex. Metaregression was not required to test the influence of EPDs on hemodynamic instability changes by dilations, because all five studies included in the meta-analysis of hemodynamic instability had 100% of their carotids protected by EPDs. Results from our meta-regression confirmed that two other mentioned factors did not contribute to the heterogeneity among studies. Also, metaregression confirmed that between-study distribution difference of EPD was not the source of heterogeneity.¹³

Sensitivity analysis. The influence of hemodynamic instabilities with and without either of two dilation techniques was analyzed, using the random effects estimate. The analysis indicated that no individual study had a disproportionate effect on the RR. Likewise, in the sensitivity analysis of the neurologic events and mortality, the pooled point estimates remained consistent across sensitivity analyses, with some variation in the magnitude of the effect size.

DISCUSSION

Our study assessed the short-term outcomes of various balloon dilation techniques on short-term outcomes after CAS. Based on data from seven nonrandomized studies, we found an absolute benefit in hemodynamic instability when postdilation was avoided during transfemoral CAS persisting, at least postoperative until discharge. We also found that neither predilation nor postdilation was associated with a mortality risk difference during the early postoperative period. Likewise, neither of the two dilation techniques had an independent effect on the risk of developing neurologic events. However, a meta-analysis of the neurologic events reported in the studies showed contradicting overall results in fixed and random effects models. Unlike the random effects model, the fixed effects model showed a significant effect of dilation techniques (regardless of their type) on the overall risk of developing neurologic events during the periprocedural period and 30 days after CAS. Clinically, this finding means that the fewer dilations performed, the better neurologic outcome is expected during the periprocedural period and 30 days after CAS. Notably, if results of smaller studies are

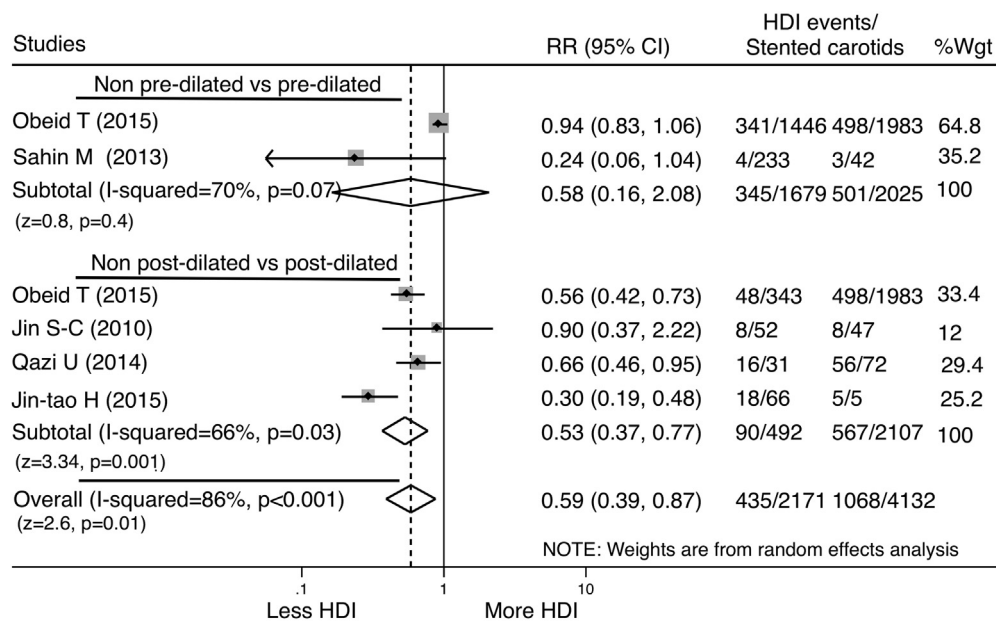


Fig 2. Forest plot of the random effects pooled hemodynamic instability (HDI) risk ratio (RR) estimate. CI, Confidence interval; Wgt, weight.

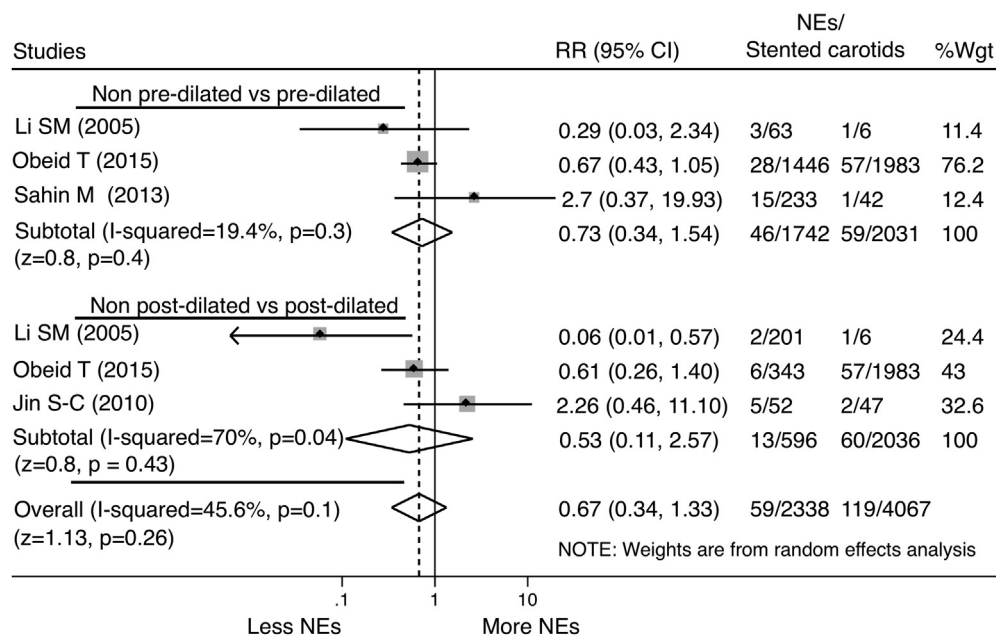


Fig 3. Forest plot of the random effects pooled neurologic events (NEs) risk ratio (RR) estimate. CI, Confidence interval; Wgt, weight.

systematically different from results of larger ones, as is observed in Figs 3 and 5, then a random effects meta-analysis would exacerbate the effects of the bias.^{10,11} This can happen as a result of publication bias or within-study bias in smaller studies. Also, it is clinically more conservative to avoid a procedure with a probable adverse effect until proven otherwise. Both statistically and clinically, accepting the overall results of the fixed

model meta-analysis for neurologic events seems to be superior.

A 2013 systematic review and meta-analysis of CAS-induced hemodynamic instability revealed a number of statistically significant associations. These included mean age with hemodynamic instability, a distance of 10 mm or less between the carotid bifurcation and the site of minimum lumen diameter with bradycardia,

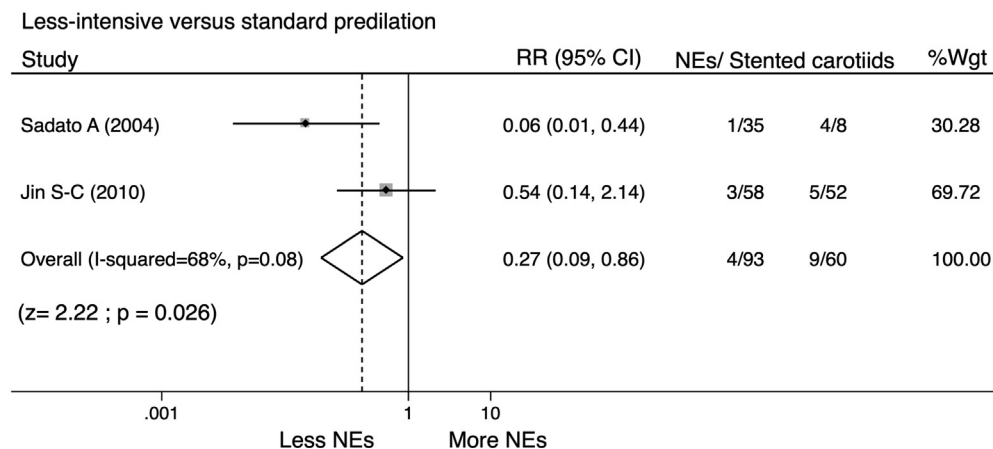


Fig 4. Forest plot of the fixed effects pooled neurologic events (NEs), comparing standard vs less intensive predilation. *CI*, Confidence interval; *RR*, risk ratio; *Wgt*, weight.

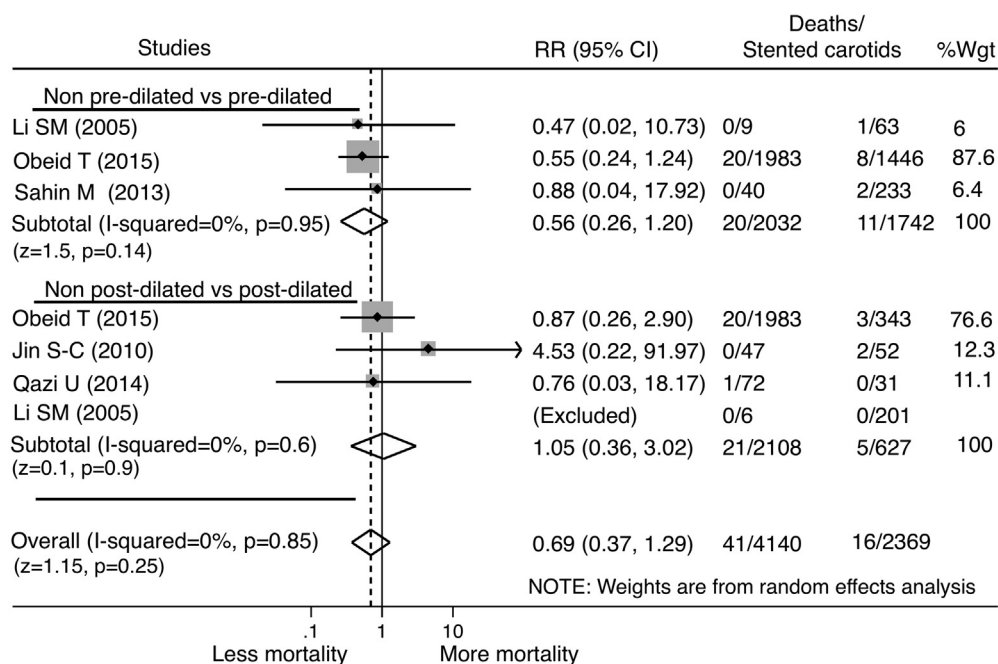


Fig 5. Forest plot of the random effects pooled mortality estimate. *CI*, Confidence interval; *RR*, risk ratio; *Wgt*, weight.

and prior ipsilateral CEA with persistent hemodynamic instability.²³ However, predilation and postdilation association with hemodynamic instability were not analyzed in their study. The impact of hemodynamic instability on the clinical outcome of the patients remains unclear. Their results did not show a statistically significant increase in death, stroke, transient ischemic attack, or myocardial infarction rates among patients with and without hemodynamic instability. However, several studies have demonstrated more extended hospital stays of patients with hemodynamic instability.^{6,24,25} Also, some authors have reported high rates of acute renal failure after CAS-induced hypotension lasting

more than 3 hours.²⁶ Therefore, we would encourage the prevention, recognition, and correction of CAS-induced hemodynamic instabilities, and our results suggest that predilation is associated with such hemodynamic instabilities.

Prolapsed debris and emboli released from an unstable plaque through the cells of the stent are assumed to be the possible source of neurologic events. However, our subgroup analysis showed that postdilation is not the only source of this theoretical neurologic events but avoiding either of two dilations might provide an overall safer operation. Some other available studies have reported the same result as we did.²⁷ They also showed

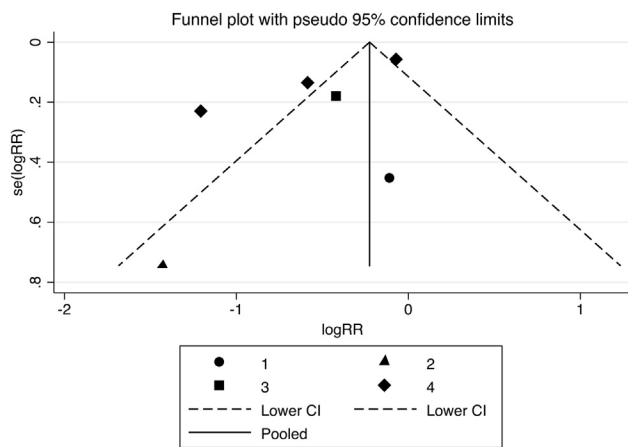


Fig 6. Publication bias test by hemodynamic instability. Symbols of similar shape and color represent predilation and postdilation impacts reported by the same study for the two subgroups. *CI*, Confidence interval; *RR*, risk ratio.

that the stent design (open cell vs closed cell) could be a significant risk factor for developing neurologic events. We did not examine stent design in this report.

In an extensive logistic regression analysis of the effect of different predilation and postdilation combination, Obeid et al¹⁵ performed a retrospective study, using the Vascular Quality Initiative database between 2005 and 2014. They concluded that postdilation increases the chances of perioperative hemodynamic instability, stroke, and mortality in patients undergoing CAS. The findings of this review regarding hemodynamic instability are consistent with previous studies. They also concluded that performing transfemoral CAS without postdilation has a better outcome. However, our findings suggest a possible rather than an absolute neurologic benefit for excluding dilation techniques. Further, this possible benefit seems to be achieved by excluding either predilation or postdilation techniques. The following facts may, in part, explain these differences. First, a short-term neurologic event is an independently defined end point in our review, whereas mortality and short-term and long-term neurologic events were defined as a single end point in their analysis. Second, their adjusted odds ratio showed a marginal significance ($P = .05$) and we preferred to report unadjusted pooled RR because the heterogeneity of confounder adjustment strategies in observational studies may lead to further bias for summary effect size in meta-analyses, especially for weak or medium associations so that the direction of causal inference would be even reversed on occasion.²⁸

Most other published studies regarding CAS dilations have been case series. Although we did find one available systematic review that had pooled the proportion of adverse effects reported by these case series for carotids either predilation, postdilation, or in

combination.²⁹ This study pooled proportions of the outcome from a set of case series of selectively postdilated carotids and compared it with the pooled proportion of the outcome from other standard dual-dilated series. However, comparing these two pooled proportions would be inappropriate, because the summary effect of each group is the pooled proportion of studies with different settings and insufficient within-study control. For example, intervention pooled effect might represent studies using EPDs whereas control pooled effect represents studies without EPDs. Also, the question remains about how outcomes may change if dilations are excluded in all procedures as opposed to avoiding dilations selectively. To avoid this selection bias and assess the effect of two commonly carried out dilations in CAS, the present review is the first systematic review and meta-analysis that only includes studies if they have both cohort and control arms.

Nowadays, most centers consider postdilation if completion angiography shows residual stenosis greater than 30%, according to the North American Symptomatic Carotid Endarterectomy Trial method. However, some centers perform postdilation if completion angiography shows greater than 40% residual stenosis.¹⁹ Likewise, and based on findings from this systematic review and meta-analysis, and assuming that the primary goal of CAS is stabilization of the plaque rather than blood flow restoration, we would recommend thresholds of residual stenosis of greater than 30% to prompt postdilation. Because there is intuitively a direct relationship between the severity of residual stenosis and the risk of future progression to severe recurrent stenosis, and possibly symptoms, the benefit of limiting angioplasty intensity or frequency needs to be balanced against this likely increased long-term risk of recurrent critical stenosis. Future studies may determine the optimal acceptable residual stenosis. Less intensive predilation models as offered by some studies, inflate the dilating balloon of less than 5 mm,^{20,21} or perform single balloon inflation for less than 6 seconds as alternative approaches to prevent unwanted adverse effects.³⁰ Passing EPD through the lesion, before any plaque angioplasty (predilation) is advised by some experts to decrease perioperative stroke,³¹ which might be especially useful in less calcified lesions and stenosis of less than 80%.³⁰ Our result suggests that using dilations with different intensities is unlikely to cause an increased risk of neurologic events in radiation-induced carotid stenoses. However, more studies are required to confirm this result obtained from a single analyzed study, because plaque stability may behave differently in these patients. Some studies indicate that radiation may even accelerate the progression of atherosclerosis.³² Still, stenotic carotid lesions in patients with previous cervical radiation are less inflammatory but more fibrotic than atherosclerotic carotid lesions in nonradiated patients.³³ In fact, the intimal

proliferation of mainly fibrous tissue and marked thickening of the endothelium leads to luminal narrowing.³⁴ Regarding hemodynamic instability, applying the appropriate prophylactic measures and strict blood pressure and heart rate monitoring during the procedure and immediately after should be encouraged for early recognition and correction of these hemodynamic disturbances.²³ In addition to prophylactic atropine, some authors suggest using temporary venous demand pacing to prevent life-threatening baroreceptor-induced hypotension.³⁵

Limitations. We are aware that the pooling of data from nonrandomized studies is a debated topic in the field of meta-analysis.^{36,37} However, previous studies were inconclusive regarding the clinical impact of carotid stenting dilation techniques. Furthermore, blinding surgeons to the intervention and randomizing patients to dilation techniques to design a randomized controlled trial are neither feasible nor ethical. Moreover, the field of comparative effectiveness research³⁸ has increasingly been used to incorporate data from observational studies to complement randomized controlled trials^{39,40} and to generate evidence that can guide health care decisions. Therefore, we aimed to provide evidence of the optimal use of dilation techniques in carotid stenting by conducting a systematic review and meta-analysis of the available studies. While conducting this systematic review, we also considered strategies to minimize the risk of confounders and bias.⁴¹

To perform the meta-analysis, we used both fixed and random effects models and discussed the preferred model wherever results were not concordant. We used metaregression test to explore the impact of the different between-study distributions of symptomatic lesions, sex, and using EPDs on the estimated effect measures. The results showed that neither of these three covariates was the source of heterogeneity between studies.¹³ As shown in Table 1, several studies had already controlled for some of the measured potential confounders by matching. To address the potential residual confounding in the included studies, we conducted a thorough sensitivity analysis^{42,43} for unmeasured confounding. No single study affected the summary estimates.

Additionally, publication bias was one of the possible sources of asymmetry in the hemodynamic instability funnel plot.⁴⁴ As such, we compared the fixed and random effects estimates of the intervention effect to investigate this observed asymmetry.⁴⁵ The outcomes were consistent in both models except in neurologic events wherein the overall summary effect was significant in the fixed effects model (Supplementary Fig 2, online only) and nonsignificant in a random model. We also discussed the reason that for this secondary outcome a fixed model could be a more conservative model to

follow, either statistically or clinically. In addition to predilation and postdilation, there are other predictors for perioperative neurologic complications. Despite all of these strategies, the impact of dilations cannot be adjusted in our analysis by predictors including baseline stenosis, the type of stenoses (primary vs recurrent after CEA/CAS), the distance between bifurcation and lesions, type and number of stents, operator level of experience, plaque length, use of statins, and the plaque calcification severity^{46,47}; therefore, additional studies are needed to investigate the adjusted effects of these factors using univariate and multivariate models. Also, further studies are required to subgroup analyze hemodynamic instabilities of different severities—for example, those requiring intervention vs mild self-limiting hemodynamic changes, or asymptomatic bradycardia. Such additional studies may clarify the situation more, considering that our meta-analysis had little power, as it was based on relatively few studies. Therefore, a definitive conclusion should be drawn when more studies become available.⁴⁸

CONCLUSIONS

Avoiding postdilation has an absolute intraoperative benefit in decreasing hemodynamic instabilities, and this benefit lasts at least 30 days. Regardless of the type, fewer dilations during CAS will possibly decrease the neurologic events during and 30 days after the procedure. We would like to suggest a less aggressive approach to postdilation after CAS.

Future studies should find the optimal threshold to prompt postdilation. Perhaps accepting residual stenosis slightly above currently practiced 30% may be more conservative. Definitive conclusions regarding the long-term effects of dilation techniques require that more studies are performed.

AUTHOR CONTRIBUTIONS

Conception and design: BZ, MS, MI, LS, ST, PS

Analysis and interpretation: BZ, PS

Data collection: BZ, PS

Writing the article: BZ, PS

Critical revision of the article: MS, MI, LS, ST, PS

Final approval of the article: BZ, MS, MI, LS, ST, PS

Statistical analysis: BZ

Obtained funding: Not applicable

Overall responsibility: PS

REFERENCES

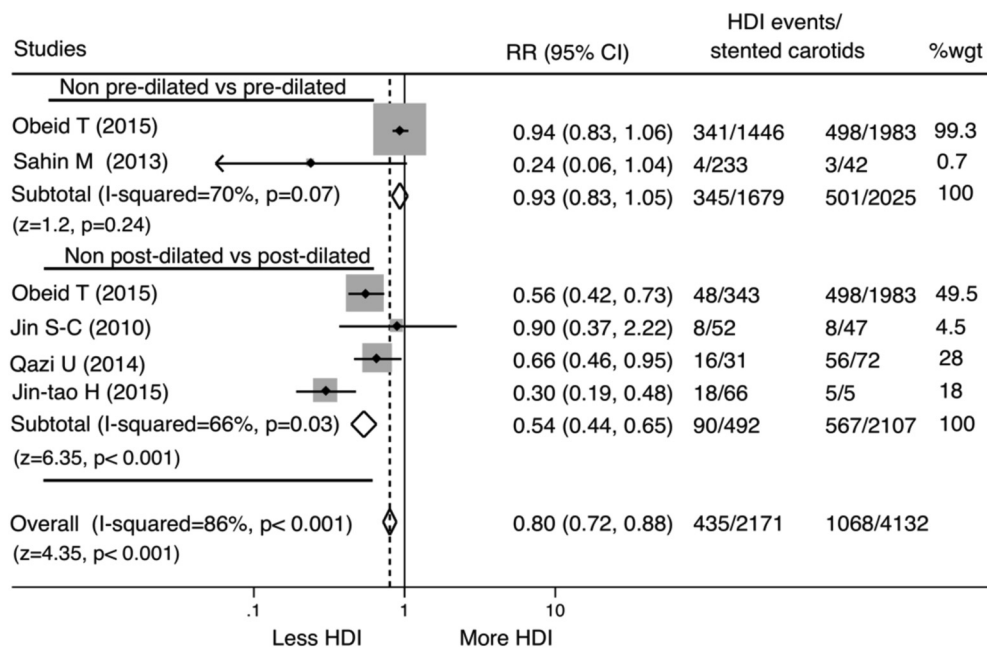
1. Cui L, Han Y, Zhang S, Liu X, Zhang J. Safety of stenting and endarterectomy for asymptomatic carotid artery stenosis: a meta-analysis of randomised controlled trials. *Eur J Vasc Endovasc Surg* 2018;55:614-24.
2. Bangalore S, Kumar S, Wetterslev J, Bavry AA, Gluud C, Cutlip DE, et al. Carotid artery stenting vs carotid endarterectomy: meta-analysis and diversity-adjusted trial sequential analysis of randomized trials. *Arch Neurol* 2011;68:172-84.

3. Brott TG, Hobson RW, Howard G, Roubin GS, Clark WM, Brooks W, et al. Stenting versus endarterectomy for treatment of carotid-artery stenosis. *N Engl J Med* 2010;363:11-23.
4. Gupta R, Abou-Chebl A, Bajzer CT, Schumacher HC, Yadav JS. Rate, predictors, and consequences of hemodynamic depression after carotid artery stenting. *J Am Coll Cardiol* 2006;47:1538-43.
5. Mlekusch W, Schillinger M, Sabeti S, Nachtmann T, Lang W, Ahmadi R, et al. Hypotension and bradycardia after elective carotid stenting: frequency and risk factors. *J Endovasc Ther* 2003;10:851-9.
6. Bussi re M, Lownie SP, Lee D, Gulka I, Leung A, Pelz DM. Hemodynamic instability during carotid artery stenting: the relative contribution of stent deployment versus balloon dilation. *J Neurosurg* 2009;110:905-12.
7. Cooper HM. Synthesizing research: a guide for literature reviews. 3rd ed. Newbury Park (CA): Sage; 1998.
8. Liberati A, Altman DG, Tetzlaff J, Mulrow C, G tzsche PC, Ioannidis JPA, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *PLoS Med* 2009;6:e1000100.
9. Higgins JPT, Thompson SC, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ* 2003;327:557-60.
10. Kjaergard LL, Villumsen J, Gluud C. Reported methodologic quality and discrepancies between large and small randomized trials in meta-analyses. *Ann Intern Med* 2001;135:982-9.
11. Poole C, Greenland S. Random-effects meta-analyses are not always conservative. *Am J Epidemiol* 1999;150:469-75.
12. Higgins JPT, Thompson SC. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002;21:1539-58.
13. Thompson SC, Higgins JPT. How should meta-regression analyses be undertaken and interpreted? *Stat Med* 2002;21:1559-73.
14. Tobias A. Assessing the influence of a single study in the meta-analysis estimate. *Stata Tech Bull* 1999;8:14-7.
15. Obeid T, Arnaoutakis DJ, Arhuidese I, Qazi U, Abularrage CJ, Black J, et al. Poststent ballooning is associated with increased periprocedural stroke and death rate in carotid artery stenting. *J Vasc Surg* 2015;62:616-22.
16. Li SM, Li D, Ling F, Miao ZR, Wang ML. Carotid artery stenting: experience of a single institute in china. *Interv Neuroradiol* 2005;11:205-12.
17. Han J, Zhao H, Li X, He Q, Ye S, Dong G, et al. [Related factors of hemodynamic damage after carotid artery stenting]. *Beijing Da Xue Xue Bao* 2015;47:804-8.
18. Sahin M, Yazicioglu MV, Acar G, Demir S, Kalkan ME, Ozkan B, et al. Safety of balloon pre-dilatation in the treatment of severe carotid artery stenosis. *Eur Rev Med Pharmacol Sci* 2013;17:788-93.
19. Qazi U, Obeid TE, Enwerem N, Schneider E, White JR, Freischlag JA, et al. The effect of ballooning following carotid stent deployment on hemodynamic stability. *J Vasc Surg* 2014;59:756-60.e1.
20. Sadato A, Satow T, Ishii A, Ohta T, Hashimoto N. Use of a large angioplasty balloon for predilation is a risk factor for embolic complications in protected carotid stenting. *Neurol Med Chir (Tokyo)* 2004;44:337-42; discussion: 343.
21. Jin SC, Kwon OK, Oh CW, Jung C, Han MG, Bae HJ, et al. A technical strategy for carotid artery stenting: suboptimal prestent balloon angioplasty without poststenting balloon dilatation. *Neurosurgery* 2010;67:1433-8.
22. Harrod-Kim P, Kadkhodayan Y, Derdeyn CP, Cross DT 3rd, Moran CJ. Outcomes of carotid angioplasty and stenting for radiation-associated stenosis. *AJNR Am J Neuroradiol* 2005;26:1781-8.
23. Mylonas SN, Moulakakis KC, Antonopoulos CN, Kakisis JD, Liapis CD. Carotid artery stenting-induced hemodynamic instability. *J Endovasc Ther* 2013;20:48-60.
24. Lin PH, Zhou W, Kougi s P, El Sayed HF, Barshes NR, Huynh TT. Factors associated with hypotension and bradycardia after carotid angioplasty and stenting. *J Vasc Surg* 2007;46:844-6.
25. Park BD, Divinagr cia T, Madej O, McPhelimy C, Piccirillo B, Dahn MS, et al. Predictors of clinically significant post-procedural hypotension after carotid endarterectomy and carotid angioplasty with stenting. *J Vasc Surg* 2009;50:526-33.
26. Taha MM, Toma N, Sakaida H, Hori K, Maeda M, Asakura F, et al. Periprocedural hemodynamic instability with carotid angioplasty and stenting. *Surg Neurol* 2008;70:276-9.
27. Doig D, Turner EL, Dobson J, Featherstone RL, Lo RTH, Gaines PA, et al. Predictors of Stroke, Myocardial Infarction or Death within 30 Days of Carotid Artery Stenting: results from the International Carotid Stenting Study. *Eur J Vasc Endovasc Surg* 2016;51:327-34.
28. Liu T, Nie X, Wu Z, Zhang Y, Feng G, Cai S, et al. Can statistic adjustment of OR minimize the potential confounding bias for meta-analysis of case-control study? A secondary data analysis. *BMC Med Res Methodol* 2017;17:179.
29. Petr O, Brinjikji W, Murad MH, Glodny B, Lanzino G. Selective-versus-standard poststent dilation for carotid artery disease: a systematic review and meta-analysis. *AJNR Am J Neuroradiol* 2017;38:999-1005.
30. Kastrup A, Skalej M, Krapf H, Nagele T, Dichgans J, Schulz JB. Early outcome of carotid angioplasty and stenting versus carotid endarterectomy in a single academic center. *Cerebrovasc Dis* 2003;15:84-9.
31. Vos JA, van den Berg JC, Ernst SM, S ttorp MJ, Overtoom TT, Mauser HW, et al. Carotid angioplasty and stent placement: comparison of transcranial Doppler US data and clinical outcome with and without filtering cerebral protection devices in 509 patients. *Radiology* 2005;234:493-9.
32. Gujral DM, Chahal N, Senior R, Harrington KJ, Nutting CM. Radiation-induced carotid artery atherosclerosis. *Radiother Oncol* 2014;110:31-8.
33. Fokkema M, Den Hartog AG, Van Lammeren GW, Bots ML, Pasterkamp G, Vink A, et al. Radiation-induced carotid stenotic lesions have a more stable phenotype than de novo atherosclerotic plaques. *Eur J Vasc Endovasc Surg* 2012;43:643-8.
34. Dorresteijn LD, Kappelle AC, Scholz NMJ, Munneke M, Scholma JT, Balm AJM, et al. Increased carotid wall thickening after radiotherapy on the neck. *Eur J Cancer* 2005;41:1026-30.
35. Harrop JS, Sharan AD, Benitez RP, Armonda R, Thomas J, Rosenwasser RH. Prevention of carotid angioplasty-induced bradycardia and hypotension with temporary venous pacemakers. *Neurosurgery* 2001;49:812-4.
36. Faber T, Ravaud P, Riveros C, Perrodeau E, Dechartres A. Meta-analyses including non-randomized studies of therapeutic interventions: a methodological review. *BMC Med Res Methodol* 2016;16:35.
37. Shapiro S. Meta-analysis/shmeta-analysis. *Am J Epidemiol* 1994;140:771-8.
38. Sox HC. Defining comparative effectiveness research: the importance of getting it right. *Med Care* 2010;48(6 Suppl):S7-8.
39. Alemayehu D, Cappelleri JC. Revisiting issues, drawbacks and opportunities with observational studies in comparative effectiveness research. *J Eval Clin Pract* 2013;19:579-83.

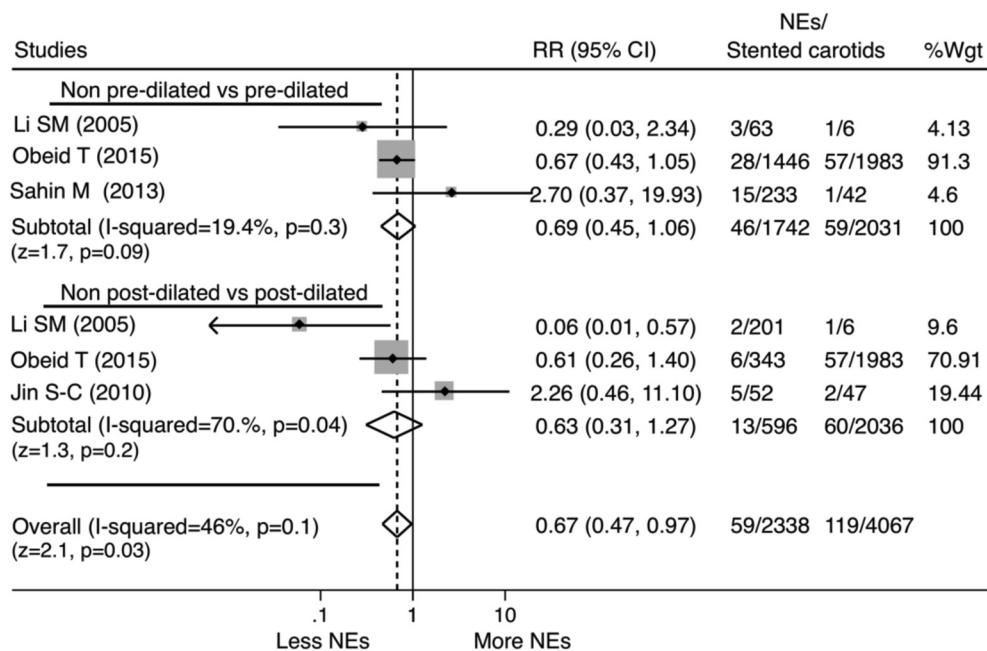
40. Norris SL, Atkins D, Bruening W, Fox S, Johnson E, Kane R, et al. Observational studies in systematic [corrected] reviews of comparative effectiveness: AHRQ and the Effective Health Care Program. *J Clin Epidemiol* 2011;64: 1178-86.
41. Navarese EP, Kozinski M, Pafundi T, Andreotti F, Buffon A, Servi SD, et al. Practical and updated guidelines on performing meta-analyses of non-randomized studies in interventional cardiology. *Cardiol J* 2011;18:3-7.
42. Mathur MB, VanderWeele TJ. Sensitivity analysis for unmeasured confounding in meta-analyses. *J Am Stat Assoc* 2019; <https://doi.org/10.1080/01621459.2018.1529598>:1-36.
43. Groenwold RH, Nelson DB, Nichol KL, Hoes AW, Hak E. Sensitivity analyses to estimate the potential impact of unmeasured confounding in causal research. *Int J Epidemiol* 2009;39:107-17.
44. Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple graphical test. *BMJ* 1997;315:629-34.
45. Sterne JA, Sutton AJ, Ioannidis JPA, Terrin N, Jones DR, Lau J, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ* 2011;343:d4002.
46. Mathur A, Roubin GS, Iyer SS, Piamsonboon C, Liu MW, Gomez CR, et al. Predictors of stroke complicating carotid artery stenting. *Circulation* 1998;97:1239-45.
47. Rizwan M, Faateh M, Dakour-Aridi H, Nejim B, Alshwaily W, Malas MB. Statins reduce mortality and failure to rescue after carotid artery stenting. *J Vasc Surg* 2019;69:112-9.
48. Greco T, Zangrillo A, Biondi-Zoccai G, Landoni G. Meta-analysis: pitfalls and hints. *Heart Lung Vessel* 2013;5:219-25.

Submitted Jul 17, 2019; accepted Nov 23, 2019.

Additional material for this article may be found online at www.jvascsurg.org.



Supplementary Fig 1 (online only). Forest plot of the fixed effects pooled hemodynamic instability (HDI) risk ratio (RR) estimate. CI, Confidence interval; wgt, weight.



Supplementary Fig 2 (online only). Forest plot of the fixed effects model pooled neurologic events (NEs) risk ratio (RR) estimate. CI, Confidence interval; Wgt, weight.